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Project Title:	MODIFICATION OF TURBULENT NON-PREMIXED FLAMES BY PULSED FUEL INJECTION		
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Program/Discipline:	COMBUSTION SCIENCE		
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No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
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Key Personnel Changes/Previous PI:			
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Task Description:	Strongly-pulsed turbulent flames are studied by the injection of a fuel jet of unheated gaseous hydrocarbon fuel (ethylene) into a co-flowing oxidizer. In all cases the flames are fully-modulated, with the fuel jet completely shut off between injection intervals. Optical diagnostics and probes are used to determine the flame length, the dynamics of the large-scale flame puff structures, the velocity field and the rate of oxidizer entrainment, the combustion temperatures, flame radiation, and the exhaust emissions of the pulsed and steady flames. Experiments to study the pulsed flame structure and thermal characteristics are conducted in reduced gravity using the NASA GRC 2.2 second Drop Tower. Laboratory experiments to examine the flame structure and exhaust emissions are performed at the UW (University of Washington); velocity field measurements are undertaken at WPI (Worcester Polytechnic Institute). The combustion characteristics of fully-modulated diffusion flames are also investigated by three-dimensional, unsteady computations using both direct numerical and large eddy simulation approaches. The overall objective of this research program is to determine the behavior of strongly-pulsed and steady turbulent diffusion flames in both normal and reduced gravity. Emphasis is placed on examining the effects of buoyancy on the development and interaction of the turbulent flame structures over a wide range of injection conditions, ranging from isolated, non-interacting flame puffs to flames with steady injection. The implications of the dynamics and interaction of the burning fuel puffs for fuel/air mixing, combustion, and exhaust emissions are to be established. Combustion is critical to many systems essential to the technical infrastructure of the United States and of the world. Examples of important applications of combustion technology range from domestic furnaces and water heaters to large-scale central power stations. Turbulent combustion, which involves the complex coupling of chemical and fluid mechanical processes, is a critical fundamental problem for combustion processes. Specifically, combustion devices that are based on unsteady, turbulent flow appear to have the potential to achieve the low levels of emissions required to meet increasingly stringent environmental regulations, as well as economical savings that might derive from the use of more compact combustors. However, the physical phenomena responsible for the enhanced mixing in unsteady, reacting flows are not currently well understood (the mechanisms operative in steady, turbulent jet flames are also not completely understood). Part of the difficulty in either case stems from the presence of buoyancy, which significantly complicates the turbulent mixing in flames in normal gravity. By the use of the microgravity environment to effectively remove the effects of buoyancy, an improved fundamental understanding of the fuel/air mixing and combustion characteristics of both fully-modulated and steady-state combustion systems can result. The resulting new fundamental knowledge may then lead to improved combustor designs for higher efficiency, more economical operation, and lower emissions for numerous applications on earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work to date suggests that there are apparently specific injection conditions where the emissions of fully-modulated flames are comparable to those of the steady flame while the flame length is significantly shorter. This points to the potential utility of this technique in the development of compact, low emissions combustors involving turbulent diffusion flames.
	Activities over the past year included: 1) pulsed-flame experiments in microgravity in the 2.2 s Drop tower at NASA Glenn Research Center; 2) measurements of pulsed-flame emissions in the laboratory, 3) surveys of turbulent puffs using Particle Image Velocimetry (PIV); and 4) numerical modeling of pulsed jet diffusion flames. The combustor contains a fuel nozzle with inner diameter of 2 mm centered in a square duct 20 × 20 cm in cross section and 52 cm in length. The fuel is pure ethylene (C2H4) and the oxidizer co-flow consists of either air (lab experiments) or 30% O2 and 70% N2 by volume (Drop Tower tests). The gaseous fuel flow is controlled by a fast-response solenoid valve actuated in an on-off fashion. A continuously-energized, heated-wire igniter situated a few millimeters downstream of the nozzle exit ensures the ignition of each fuel puff upon injection. Experiments have been performed with a nozzle exit Reynolds number of 5000 and injection times ranging from 6 ms to 303 ms with jet off-times from 5 ms to 300 ms. Flame imaging is performed using a CCD array at a framing rate of 400 Hz. The mean flame lengths are determined using NASA's Spotlight image tracking program. The celerity (bulk velocity) of the turbulent flame puffs is determined by cross-correlating the temperature signals between a pair of type-R thermocouples at the location of the visible flame tip. Measurement of exhaust emissions are performed within the combustor and at the combustor exit using an aspirated sampling probe and standard emissions analyzers. Reducing the jet-off time, for a fixed injection time, leads to a significant increase in the degree of flame puff interaction and results in an increase in the flame length. The high-speed imaging results reveals details of the flame structure and the onset of flame puff interaction as the jet off-time is decreased. The structure of fully-modulated flames is seen to be generally similar for normal gravity and microgravity, with the microgravity flame puffs persisting for a slightly longer time and advecting more slowly downstream than their normal-gravity counterparts. The measured puff celerity varied from a maximum value of c = 3.2 m/s to a minimum of c = 0.86 m/s (the injection velocity was 22 m/s). Three trends are apparent: 1) an increase in the puff celerity as the jet off-time is decreased, 2) a higher celerity for a lower value of the injection time, and 3) a higher celerity in normal gravity than in microgravity. The puff celerity in normal gravity exceeds that in microgravity by as much as 40% for the shortest injection time; a similar increase was recorded for both gravity levels as the injection time parameter decreased from 46 ms to 6 ms. The celerity of the most strongly-interacting flame puffs was as much as 2.8 times greater than for the case of isolated flame puffs. This increase in celerity is consistent with a decreased rate of momentum exchange, per amount of injected fuel, between the flame puffs and the oxidizer co-flow. Although buoyancy results in only a modest change in the flame length (6-20%), there is a consistently positive effect of buoyancy on the celerity of the flame puff structures. For these strongly-pulsed flames, the values of the CO emission index (EICO) are generally higher than that of the steady flame. For sufficiently long injection time the EICO becomes comparable to the steady flame value for all values of the jet-off time. The EICO is highest for flame puffs with the shortest injection time. The observed decrease in the emission index of CO with decreasing jet-off time is consistent with the increased structure/structure interactions, which result in a longer flame length and a longer flame residence time due to a decreased rate of air entrainment per amount of fuel injected. For most injection conditions the measured exhaust-point NO concentrations are below those of the steady flame, and become comparable to the steady flame values for sufficiently short jet off-time. The increase in NO level with decreasing jet-off time is most pronounced for short injection times. The increase in the NO emission index (EINO) for short off-times is consistent with the greater flame puff interaction which leads to a slower rate of excess air dilution for a given amount of injected fuel, leading in turn to a longer flame length and a higher residence time. The emission index of CO correlates well with a global residence time based on the observed variation of flame length.

Task Progress:	The EINO by contrast does not correlate with residence time for all values of injection time. The EINO does increase with increasing residence time for a fixed value of injection time. Temperature changes that accompany changes in the rate of air entrainment per volume of injected fuel could explain why the amount of NO emitted per mass of fuel injected is strongly dependent both the flame structure as well as the residence time. There are injection time and jet-off time combinations where the emissions of fully-modulated flames are comparable to those of steady flames, but with a shorter flame length. These results point toward potential benefits of the fully-modulated injection technique in the development of compact, low emissions combustors involving turbulent diffusion flames. PIV investigations of strongly-pulsed, non-reacting jets have been conducted in a similar configuration as the combustion experiments. In this case the chamber was 30 × 30 cm in cross section and the nozzle diameter was 5 mm. Puffs were generated by injecting room temperature air through the nozzle into a flow chamber with a seeded, weak air co-flow. The seed particles were either 40 micron or 9 micron diameter aluminum oxide particles. The injection conditions were similar to the combustion experiments. Isolated jet puffs were examined at a location roughly 50 jet diameters downstream of the injection nozzle, at which location the puffs mixed sufficiently with the co-flow so that there is sufficient seeding within the puff for PIV data acquisition. The flow field was illuminated by a laser sheet from a double-pulsed Nd:YAG laser at a wavelength of 532 nm and a pulse energy of approximately 10 mJ/pulse. Images were recorded by a progressive scan CCD camera with a 60 Hz frame rate and a 640 × 480 pixel resolution. The time separation between successive pulses was on the order of 0.5 ms to 1 ms. The field of view for the velocity measurements was approximately 34 jet nozzle diameters in the radial direction and 25 diameters in the axial direction. Conditional averaging methods were developed and utilized to remove the spatial jitter resulting from the turbulent nature of the flow. The location of maximum velocity within each isolated puff used to align the velocity fields from different runs. The position of peak vorticity within the primary vortex core of the puff was used to align the vorticity fields. The vorticity measured by PIV is confined primarily within the axially-elongated vortex cores. The peak vorticity occurs approximately 5 jet diameters from the centerline at a downstream distance of 50 diameters. Relatively little vorticity resides in the wake of the vortex core. Conditional averaging significantly smoothes the average vorticity, yielding significantly lower levels of maximum vorticity than observed in the instantaneous case. The rms of the vorticity fluctuations appears to be is mainly limited to the vortex core region, and can reach as much as 50% of the peak values within the core. The turbulent shear stress distribution is quite complex for this flow field. The axial velocity fluctuation has a ring type structure similar to the vorticity field; the radial velocity fluctuation is more elongated in the axial direction. An existing computer code for steady jet diffusion flames has been modified to simulate strongly-pulsed flames. This code employs Direct Numerical Simulation (DNS) using a low Mach number approximation applied to the compressible Navier-Stokes equations, combined with a single-step, Arrehenius-type reaction scheme. A massively parallel implementation of this code has been developed and run on the Cray X1 at the Arctic Region Supercomputer Center. Pulsed jet simulations have been performed for a Reynolds number of 1000 and an injection time of 0.6 ms. This very short injection time results in a "micro-puff" configuration, which exhibits the puff-like structure of these flames. The computational domain is being extended to simulate flames over the parameter range corresponding to the experiments. This implies significantly increased numerical resolution, a large computational domain, and calculations over longer time scales. Currently, two approaches are being pursued for this purpose. The first involves implementing a non-uniform mesh, which will enable the computational resolution to be concentrated only where it is needed. The second candidate approach is to apply monotonically integrated large eddy simulation (MILES).

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